Experiment – 06

Name: Smyle Goyal

Section: 24AIT_KRG-1_A

UID: 24BAI70469

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Branch: BE-CSE(AIML)

Subject: Advanced Database Management System

Question 1:

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the di erence between the number of apples sold and the number of oranges sold on the same day. The di erence should be calculated as: Di erence = Apples Sold - Oranges Sold Display the sale date and the calculated di erence. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output:

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20

Solution:

```
SELECT
   SALE_DATE,
   SUM(
      CASE
         WHEN FRUIT = 'apples' THEN SOLD_NUM
         ELSE -SOLD_NUM
      END
   ) AS DIFF
FROM SALES AS S
GROUP BY SALE_DATE
ORDER BY SALE_DATE;
```

Question 2:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where: • id represents the current node. • p_id represents the parent node. Classify every node into one of the following categories: •

Root: A node that does not have any parent. • Leaf: A node that has a parent but does not have any children. • Inner: A node that has both a parent and at least one child. Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Solution:

```
SELECT
   ID,
   CASE
      WHEN P_ID IS NULL THEN 'ROOT'
      WHEN ID NOT IN (
         SELECT P_ID
         FROM TREE
         WHERE P_ID IS NOT NULL
      ) THEN 'LEAF'
      ELSE 'INNER'
   END AS TYPE
FROM TREE
ORDER BY ID;
```